Lecture Scribe CSE400

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Below is a faithful **lecture scribe for Lecture 8 (Gaussian Random Variables and Standard Functions)** written strictly from the provided lecture material, preserving order, definitions, formulas, derivations, identities, and the worked example exactly as presented in the slides.

All content is derived only from the lecture PDF

CSE400 – Fundamentals of Probability in Computing

Lecture 8: Gaussian, Uniform, Exponential, and Gamma Random Variables

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Outline of the Lecture

The lecture covers:

- Gaussian Random Variable
  - Definition and properties
  - Standard forms: Error function, complementary error function
  - Φ-function and Q-function

- Relation between Φ and Q
    - Example
- Types of Continuous Random Variables
- Uniform Random Variable
- Exponential Random Variable
- Laplace Random Variable
- Gamma Random Variable
    - Graph and special cases
    - Example
- Homework problem
- Problem solving
- In-class activity: Gaussian density estimation

Moments and Central Moments of a Random Variable

For a continuous random variable X:

Central Moment of Order n:

$$E[(X - \mu_X)^n] = \int (x - \mu_X)^n f_X(x)\, dx$$

For discrete case:

$$E[(X - \mu_X)^n] = \sum_k (x_k - \mu_X)^n P_X(x_k)$$

Specific Cases

Case $n = 0$:

$$E[(X - \mu_X)^0] = E[1] = 1$$

Case $n = 1$:

$$E[(X - \mu_X)] = E[X] - \mu_X = \mu_X - \mu_X = 0$$

Case $n = 2$: (Variance)

$$\sigma_X^2 = E[(X - \mu_X)^2]$$

Expanding:

$$= E[X^2 - 2\mu_X X + \mu_X^2]$$

$$= E[X^2] - 2\mu_X E[X] + \mu_X^2$$

$$= E[X^2] - \mu_X^2$$

Higher Order Moments

Skewness (n = 3):

$$C_s = \frac{E[(X - \mu_X)^3]}{\sigma_X^3}$$

- Positive → PDF is right skewed
- Negative → PDF is left skewed

Kurtosis (n = 4):

$$C_k = E[(X - \mu_X)^4]$$

A large value implies the random variable has a large peak near the mean.

Linearity Properties of Expectation

For constants a and b:

$$E[aX + b] = aE[X] + b$$

For a function expressed as a sum:

$$g(x) = g_1(x) + g_2(x) + \cdots + g_N(x)$$

Then:

$$E\left[\sum_{k=1}^{N} g_k(X)\right] = \sum_{k=1}^{N} E[g_k(X)]$$

Also:

$$E[X] = \int \quad x f_X(x)\, dx$$

Gaussian Random Variable

Definition

A Gaussian random variable is one whose PDF is given by:

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-m)^2}{2\sigma^2}\right)$$

Where:

- $m = \mu_X$ is the mean
- σ^2 is the variance

This is written as:

$$X \sim \mathrm{N}\left(m, \sigma^2\right)$$

Standard Forms

Error Function

$$\mathrm{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2}\, dt$$

Complementary Error Function

$$\mathrm{erfc}(x) = 1 - \mathrm{erf}(x)$$
$$= \frac{2}{\sqrt{\pi}} \int_x^\infty e^{-t^2}\, dt$$

Φ-function (CDF of Standard Normal)

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} \exp\left(-\frac{t^2}{2}\right) dt$$

Q-function (Gaussian Tail Function)

$$Q(x) = \frac{1}{\sqrt{2\pi}} \int_{x}^{\infty} \exp\left(-\frac{t^2}{2}\right) dt$$

Note:

$$Q(x) = 1 - \Phi(x)$$

Relation Between Φ-function and Q-function

Evaluating the CDF of a Gaussian RV

Start from:

$$F_X(x) = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y-m)^2}{2\sigma^2}\right) dy$$

Substitute:

$$t = \frac{y-m}{\sigma}$$

Then:

$$F_X(x) = \int_{-\infty}^{\frac{x-m}{\sigma}} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{t^2}{2}\right) dt$$

Hence:

$$F_X(x) = \Phi\left(\frac{x-m}{\sigma}\right)$$

Evaluating Tail Probabilities

$$P(X > x) = \int_{\frac{x-m}{\sigma}}^{\infty} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{t^2}{2}\right) dt$$

Thus:

$$P(X > x) = Q\left(\frac{x - m}{\sigma}\right)$$

Key Identities

$$Q(x) = 1 - \Phi(x)$$

$$F_X(x) = 1 - Q\left(\frac{x - m}{\sigma}\right)$$

Remarks from Slides

- To evaluate the CDF of a Gaussian RV, evaluate the Φ-function at the point.
- The Q-function is more natural for evaluating probabilities of the form $P(X > x)$.

Example

A random variable X has PDF:

$$f_X(x) = \frac{1}{\sqrt{8\pi}} \exp\left(-\frac{(x+3)^2}{8}\right)$$

Step 1: Compare with Gaussian PDF

Standard form:

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-m)^2}{2\sigma^2}\right)$$

Matching terms:

$$m = -3$$

$$\sigma^2 = 4 \Rightarrow \sigma = 2$$

Gaussian Relations Used

$$F_X(x) = \Phi\left(\frac{x-m}{\sigma}\right)$$

$$P(X > x) = Q\left(\frac{x-m}{\sigma}\right)$$

Probabilities to Find (expressed using Q-function):

1. $P(X \leqslant 0)$
2. $P(X > 4)$
3. $P(|X+3| < 2)$
4. $P(|X-2| > 1)$

Hint Given in Slides:

- Use **Q(x)** for right-tail probabilities
- Use **Φ(x)** for left-tail probabilities

This completes the lecture scribe strictly following the content, equations, derivations, and structure of the lecture PDF.

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CSE400 – Fundamentals of Probability in Computing

Lecture 8: Gaussian, Uniform, Exponential, and Gamma Random Variables

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Lecture Outline

- Gaussian Random Variable
  - Definition and properties
  - Standard forms: error function and complementary error function
  - Φ-function and Q-function
  - Relation between Φ and Q
  - Example
- Types of continuous random variables
- Uniform random variable
- Exponential random variable
- Laplace random variable
- Gamma random variable
  - Graph and special cases
  - Example
- Homework problem
- Problem solving

- In-class activity: Gaussian density estimation

Moments and Central Moments

For a continuous random variable X:

$$E[(X-\mu_X)^n] = \int (x-\mu_X)^n f_X(x)\,dx$$

For discrete case:

$$E[(X-\mu_X)^n] = \sum_k (x_k-\mu_X)^n P_X(x_k)$$

Case $n = 0$

$$E[(X-\mu_X)^0] = E[1] = 1$$

Case $n = 1$

$$E[(X-\mu_X)] = E[X] - \mu_X$$

$$= \mu_X - \mu_X = 0$$

Case $n = 2$ (Variance)

$$\sigma_X^2 = E[(X-\mu_X)^2]$$

Expand:

$$= E[X^2 - 2\mu_X X + \mu_X^2]$$

$$= E[X^2] - 2\mu_X E[X] + \mu_X^2$$

$$= E[X^2] - \mu_X^2$$

Skewness ($n = 3$)

$$C_s = \frac{E[(X - \mu_X)^3]}{\sigma_X^3}$$

- Positive → right skewed PDF
- Negative → left skewed PDF

Kurtosis ($n = 4$)

$$C_k = E[(X - \mu_X)^4]$$

Large value indicates large peak near the mean.

Linearity of Expectation

For constants a, b:

$$E[aX + b] = aE[X] + b$$

For sum of functions:

$$E\left[\sum_{k=1}^{N} g_k(X)\right] = \sum_{k=1}^{N} E[g_k(X)]$$

Also:

$$E[X] = \int x f_X(x)\, dx$$

Gaussian Random Variable

Definition

A Gaussian random variable has PDF:

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - m)^2}{2\sigma^2}\right)$$

where

- $m = \mu_X$ (mean)

- σ^2 (variance)

Notation:

$$X \sim \mathrm{N}(m, \sigma^2)$$

Standard Forms

Error Function

$$\mathrm{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt$$

Complementary Error Function

$$\mathrm{erfc}(x) = 1 - \mathrm{erf}(x)$$
$$= \frac{2}{\sqrt{\pi}} \int_x^\infty e^{-t^2} dt$$

Φ-function (Standard Normal CDF)

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x \exp\left(-\frac{t^2}{2}\right) dt$$

Q-function (Gaussian Tail)

$$Q(x) = \frac{1}{\sqrt{2\pi}} \int_x^\infty \exp\left(-\frac{t^2}{2}\right) dt$$

Identity:

$$Q(x) = 1 - \Phi(x)$$

Relation Between Φ-function and Q-function

Evaluating the CDF

$$F_X(x) = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y-m)^2}{2\sigma^2}\right) dy$$

Substitute:

$$t = \frac{y-m}{\sigma}$$

Then:

$$F_X(x) = \int_{-\infty}^{\frac{x-m}{\sigma}} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{t^2}{2}\right) dt$$

Hence:

$$F_X(x) = \Phi\left(\frac{x-m}{\sigma}\right)$$

Evaluating Tail Probability

$$P(X > x) = \int_{\frac{x-m}{\sigma}}^{\infty} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{t^2}{2}\right) dt$$

$$P(X > x) = Q\left(\frac{x-m}{\sigma}\right)$$

Key Identities

$$Q(x) = 1 - \Phi(x)$$

$$F_X(x) = 1 - Q\left(\frac{x-m}{\sigma}\right)$$

Slide Remarks

- CDF is evaluated using Φ-function
- Right-tail probabilities naturally use Q-function

Example

Given:

$$f_X(x) = \frac{1}{\sqrt{8\pi}} \exp\left(-\frac{(x+3)^2}{8}\right)$$

Compare with Gaussian PDF

Standard form:

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-m)^2}{2\sigma^2}\right)$$

Identify:

$$m = -3$$

$$\sigma^2 = 4 \Rightarrow \sigma = 2$$

Gaussian Relations Used

$$F_X(x) = \Phi\left(\frac{x-m}{\sigma}\right)$$

$$P(X > x) = Q\left(\frac{x-m}{\sigma}\right)$$

Find:

1. $P(X \leqslant 0)$
2. $P(X > 4)$
3. $P(|X+3| < 2)$
4. $P(|X-2| > 1)$

Hint from Slides

- Use **Q(x)** for right-tail probabilities
- Use **Φ(x)** for left-tail probabilities

✅ Order preserved
✅ All formulas and derivations shown step-by-step
✅ No added theory or simplifications
✅ Strictly from lecture PDF

If you'd like, I can now:

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All definitions, identities, and derivations are written **step-by-step exactly as presented**.

Source: Lecture PDF

CSE400 – Fundamentals of Probability in Computing

Lecture 8: Gaussian, Uniform, Exponential, and Gamma Random Variables

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Outline

- Gaussian Random Variable
  - Definition and Properties
  - Standard Forms: Q-function and Error Function
  - Relation between Φ-function and Q-function
  - Example
- Types of Continuous Random Variable
- Uniform Random Variable
- Exponential Random Variable
- Laplace Random Variable
- Gamma Random Variable
  - Graph and Special Cases
  - Example
- Homework Problem
- Problem Solving
- In-Class Activity: Gaussian Density Estimation

Moments and Central Moments

For a continuous random variable X:

$$E[(X-\mu_X)^n] = \int (x-\mu_X)^n f_X(x)\,dx$$

For a discrete random variable:

$$E[(X-\mu_X)^n] = \sum_k (x_k-\mu_X)^n P_X(x_k)$$

Case $n = 0$

$$E[(X-\mu_X)^0] = E[1] = 1$$

Case $n = 1$

$$E[(X - \mu_X)] = E[X] - \mu_X$$
$$= \mu_X - \mu_X = 0$$

Case $n = 2$ (Variance)

$$\sigma_X^2 = E[(X - \mu_X)^2]$$

Expand:

$$= E[X^2 - 2\mu_X X + \mu_X^2]$$
$$= E[X^2] - 2\mu_X E[X] + \mu_X^2$$
$$= E[X^2] - \mu_X^2$$

Case $n = 3$: Skewness

$$C_s = \frac{E[(X - \mu_X)^3]}{\sigma_X^3}$$

- $C_s > 0$: PDF is right skewed
- $C_s < 0$: PDF is left skewed

Case $n = 4$: Kurtosis

$$C_k = E[(X - \mu_X)^4]$$

Large value implies a large peak near the mean.

Linearity of Expectation

For constants a and b:

$$E[aX + b] = aE[X] + b$$

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Gaussian Random Variable

Definition

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where

- $m = \mu_X$
- σ^2 is the variance

Notation:

$$X \sim \mathrm{N}(m, \sigma^2)$$

Standard Forms

Error Function

$$\mathrm{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2}\, dt$$

Complementary Error Function

$$\mathrm{erfc}(x) = 1 - \mathrm{erf}(x)$$

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Φ-function (CDF of Standard Normal)

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Q-function (Gaussian Tail Function)

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Identity

$$Q(x) = 1 - \Phi(x)$$

Relation Between Φ-function and Q-function

Evaluating the CDF

$$F_X(x) = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y-m)^2}{2\sigma^2}\right) dy$$

Substitute:

$$t = \frac{y-m}{\sigma}$$

Then:

$$F_X(x) = \int_{-\infty}^{\frac{x-m}{\sigma}} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{t^2}{2}\right) dt$$

$$F_X(x) = \Phi\left(\frac{x-m}{\sigma}\right)$$

Evaluating Tail Probability

$$P(X > x) = \int_{\frac{x-m}{\sigma}}^{\infty} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{t^2}{2}\right) dt$$

$$P(X > x) = Q\left(\frac{x-m}{\sigma}\right)$$

Key Identities

$$Q(x) = 1 - \Phi(x)$$

$$F_X(x) = 1 - Q\left(\frac{x-m}{\sigma}\right)$$

Notes from Slides

- To evaluate the CDF of a Gaussian RV, evaluate the Φ-function at the point.
- The Q-function is more natural for evaluating probabilities of the form $P(X > x)$.

Example

Given:

$$f_X(x) = \frac{1}{\sqrt{8\pi}} \exp\left(-\frac{(x+3)^2}{8}\right)$$

Comparison with Gaussian PDF

Standard Gaussian PDF:

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-m)^2}{2\sigma^2}\right)$$

Identifying parameters:

$$m = -3$$

$$\sigma^2 = 4 \Rightarrow \sigma = 2$$

Gaussian Relations Used

$$F_X(x) = \Phi\left(\frac{x-m}{\sigma}\right)$$

$$P(X > x) = Q\left(\frac{x-m}{\sigma}\right)$$

Required Probabilities

1. $P(X \leqslant 0)$
2. $P(X > 4)$
3. $P(|X+3| < 2)$
4. $P(|X-2| > 1)$

Hint (as given)

- Use **Q(x)** for right-tail probabilities
- Use **Φ(x)** for left-tail probabilities

This scribe **exactly mirrors the lecture PDF**, is **exam-oriented**, and contains **no additional theory or interpretation** beyond the provided material.

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